

COMMUNICATIONS TECHNOLOGY

Communications technology affects all aspects of our lives in a fundamental way. Having an understanding of communications technology is an important part of being both technologically and media literate.

Communications technology courses are project-based and will provide students with opportunities to acquire the knowledge and skills required to design, use, and manage electronic, live, recorded, and graphic communications systems, specifically in the areas of TV, video, and movie production; radio and audio production; print and graphic communications; photography; digital imaging; broadcast journalism; and interactive new media and animation. These courses will help students understand the effects of communications technology on the environment and society. Students will also examine standards and regulations governing communications technology, health and safety issues, careers in the field, and the importance of lifelong learning, and will learn about the Essential Skills and work habits that are important for success in careers in the field.

The list of approved emphasis areas for communications technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Communications Technology: Broadcast and Print Production, Grade 11

Open

TGJ30

This course enables students to develop knowledge and skills in the areas of graphic communication, printing and publishing, audio and video production, and broadcast journalism. Students will work both independently and as part of a production team to design and produce media products in a project-driven environment. Practical projects may include the making of signs, yearbooks, video and/or audio productions, newscasts, and documentaries. Students will also develop an awareness of related environmental and societal issues, and will explore secondary and postsecondary education and training pathways and career opportunities in the various communications technology fields.

Prerequisite: None

A. COMMUNICATIONS TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of the core concepts, techniques, and skills required to produce a range of audio, video, broadcast journalism, graphic arts, and printing and publishing products or services;
- A2.** demonstrate an understanding of technical terminology, scientific concepts, and mathematical concepts used in audio and video production, broadcast journalism, graphic arts, and printing and publishing, and apply them to the creation of media products;
- A3.** demonstrate an understanding of and apply the interpersonal skills necessary to work effectively in a team setting.

SPECIFIC EXPECTATIONS

A1. Core Concepts, Techniques, and Skills

By the end of this course, students will:

- A1.1** demonstrate an understanding of technological concepts (*e.g., aesthetics, control, environmental sustainability, ergonomics, fabrication/building/creation, function, innovation, material, mechanism, power and energy, safety, structure, systems*) and their relevance to the design and creation of media projects (see pp. 7–8);
- A1.2** use audio, video, broadcast journalism, graphic arts, and publishing equipment safely and correctly to perform basic production tasks or create simple products (*e.g., shoot footage with a video camera, set up microphones, operate a sound board, print a pamphlet on various printing devices*);
- A1.3** use video editing, audio editing, graphic arts, and publishing software correctly to perform basic production tasks or create simple products (*e.g., use dissolves and fades, stop motion; add titles; create simple layouts, logos*);
- A1.4** demonstrate an understanding of the creative skills and techniques required to produce effective video, audio, broadcast journalism, graphic arts, and print products (*e.g., lighting techniques, recording techniques, blocking techniques, editing techniques, interviewing and journalistic writing techniques, colour matching*);

A1.5 identify the components of audio, video, graphic arts, and printing systems (*e.g., video cameras, editing software, switchers, microphones, audio mixers, computers, monitors, platesetters, different types of presses [offset, flexographic, gravure, letterpress], inkjet and electrostatic printers, bindery equipment*) and devices (*e.g., camera controls, sensor, recording media, connectors, toner, roller, ozone filter*), and describe their functions;

A1.6 use safe and appropriate methods for installing, transporting, and storing audio, video, broadcast journalism, and graphic arts equipment (*e.g., cameras, tripods, lights, microphones, teleprompters, portable mixers, cables, scanners*) and setting up printing equipment;

A1.7 apply design principles (*e.g., balance, rhythm, proportion, contrast, and flow*) and elements (*e.g., colour, line, space, form, and texture*) to communicate an idea or concept.

A2. Technical Terminology and Scientific and Mathematical Concepts

By the end of this course, students will:

- A2.1** demonstrate an understanding of terminology related to video and audio production, broadcast journalism, graphic arts, and publishing, and use it correctly in oral and written communication (*e.g., sampling rate, aspect ratio, headroom, continuity, offline media, closure, voice-over, tag, spot colour, colour mode*);

A2.2 demonstrate an understanding of scientific concepts that relate to processes and technologies used in video and audio production and printing (e.g., *behaviour of light; conversion of light and sound to electricity and back; digital encoding of light, sound, and data; persistence of vision; principles of various printing technologies such as inkjet, electrostatic, and offset*);

A2.3 use mathematical concepts and formulas as required to complete tasks in audio and video production, graphic arts, and print production (e.g., *calculation of lighting ratios and exposures, timing of sequences in audio and video editing, image scaling, imposition planning*).

A3. Teamwork

By the end of this course, students will:

A3.1 explain the value of sharing ideas, information, resources, and expertise when working in a team setting;

A3.2 describe and use techniques that encourage participation by all members of a team (e.g., *brainstorming, group discussion, celebration of others' thoughts or contributions*);

A3.3 describe and use concepts and techniques that facilitate effective collaboration in a team environment (e.g., *cooperative discussion, conflict resolution techniques, motivation techniques, respect for the ideas of others, constructive criticism*).

B. COMMUNICATIONS TECHNOLOGY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** apply project management techniques to the planning and development of audio, video, broadcast journalism, graphic arts, and printing products;
- B2.** apply a design process or other problem-solving processes or strategies to meet a range of challenges in creating audio, video, broadcast journalism, graphic arts, and printing products;
- B3.** create products or productions that demonstrate competence in the application of creative and technical skills.

SPECIFIC EXPECTATIONS

B1. Project Management

By the end of this course, students will:

- B1.1** use a variety of planning techniques and tools (*e.g., research, project proposals, storyboards, script treatments*) when creating plans for an audio, video, broadcast journalism, graphic arts, and/or printing project;
- B1.2** use appropriate organizational and time-management tools (*e.g., student planners, journals, production schedules, electronic organizers, organizational software*) throughout the project to manage resources and ensure that project deadlines are met.

B2. Problem Solving

By the end of this course, students will:

- B2.1** apply the steps in a design process to develop solutions to creative challenges (*e.g., define the problem or challenge, taking into account relevant contextual or background information; define project objectives and criteria and identify constraints such as cost, time, or technology limitations; gather information and generate possible solutions using techniques such as brainstorming; evaluate possible solutions and apply the one that most effectively meets the objectives and criteria within the existing constraints*) (see pp. 22–23);

- B2.2** apply appropriate problem-solving approaches and/or techniques to solve specific technical problems (*e.g., diagnostics, reverse engineering, trial and error, divide and conquer, parts substitution, extreme cases*).

B3. Process and Production Skills

By the end of this course, students will:

- B3.1** apply creative skills, equipment operating skills, and software skills to create components for a media production (*e.g., news copy, video footage, voice-overs for a TV news broadcast; text, images, page layouts for a publication*);
- B3.2** apply editing skills to integrate the components into a unified and effective production.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** describe the impact of communications media technologies and activities on the environment, and identify ways of reducing their harmful effects;
- C2.** demonstrate an understanding of social effects and issues arising from the use of communications media technologies and the importance of respecting cultural and societal diversity in the production of media projects.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** describe the effects of current audio, video, broadcast journalism, graphic arts, and printing technologies on the environment (*e.g., increased energy consumption, waste and disposal problems created by rapid obsolescence, toxic wastes, noise pollution, electromagnetic interference, RF pollution*);
- C1.2** describe ways in which environmental problems are being or can be addressed by the audio, video, broadcast journalism, graphic arts, and printing industries (*e.g., using energy-efficient equipment, upgrading rather than replacing obsolete equipment, recycling equipment slated for disposal, using environmentally friendly inks and environmentally responsible press cleanup methods, using the persuasive power of the media to promote environmental stewardship, environmental certification of operations [EcoLogo, ISO 14001]*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** demonstrate an understanding of social standards and cultural sensitivity and use appropriate and inclusive content, images, and language in communications media productions (*e.g., including people from different races, cultures, and backgrounds in media productions; portraying minority groups with respect and sensitivity; avoiding sexism, homophobia, and cultural or racial bias*);
- C2.2** identify legal and ethical issues related to communications media production (*e.g., copy-right, respect of privacy and personal information*);
- C2.3** identify recent innovations in audio, video, broadcast journalism, graphic arts, and printing technologies (*e.g., increasing affordability and ease of use of high-quality equipment; development of software for performing complex tasks*) and describe their social and economic effects (*e.g., new opportunities for small businesses to produce CDs, videos, and other products; weakening of the traditional newspaper and magazine market; new opportunities to reach specialized audiences through short-run publishing; new opportunities for Aboriginal or other ethnocultural or social groups to promote their culture and points of view; decline of old technology businesses and rise of new technology businesses*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and apply safe work practices in the use of audio, video, broadcast journalism, graphic arts, and printing equipment;
- D2.** identify careers in audio and video production, broadcast journalism, graphic arts, and printing and publishing, and describe the skills, work habits, education, and training required for entry into employment in these fields.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** describe industry hazards (*e.g., ergonomic hazards, electrical hazards, mechanical hazards*), identify sources of hazard information (*e.g., Workplace Hazardous Materials Information System [WHMIS], Passport to Safety*), and describe methods for preventing accidents;
- D1.2** demonstrate an understanding of and apply safe work practices when using equipment (*e.g., use of gloves and other protective clothing, correct seat placement, use of proper cable management techniques, use of ergonomically designed equipment, proper grounding of electrical devices, use of safety chains, correct use of ladders, proper use and storage of equipment*);
- D1.3** describe health risks (*e.g., carpal tunnel syndrome, eye strain*) associated with the use of audio, video, broadcast journalism, graphic arts, and printing equipment, and identify ways of avoiding them.

D2. Career Opportunities

By the end of this course, students will:

- D2.1** identify career opportunities in audio and video production, broadcast journalism, graphic arts, and printing and publishing (*e.g., director, announcer, videographer, journalist, graphic illustrator, printer*) and describe the qualifications and skills needed for entry into these occupations;
- D2.2** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the audio and video production, broadcast journalism, graphic arts, and printing and publishing industries (*e.g., mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);
- D2.3** demonstrate an understanding of and apply the Essential Skills that are important for success in the audio and video production, broadcast journalism, graphic arts, and printing and publishing industries, as identified in the Ontario Skills Passport (*e.g., computer use, oral communication, measurement and calculation, numerical estimation, job task planning and organizing, decision making*);
- D2.4** demonstrate an understanding of and apply the work habits that are important for success in the audio and video production, broadcast journalism, graphic arts, and printing and publishing industries, as identified in the Ontario Skills Passport (*e.g., working safely, teamwork, reliability, initiative, customer service, entrepreneurship*);
- D2.5** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in audio and video production, broadcast journalism, graphic arts, and/or printing and publishing (*e.g., work logs, skills checklist, photographs, digital media, sketches, drawings*), and explain why having a current portfolio is important for career development and advancement.